

Botox (onabotulinumtoxinA) in cervical dystonia

Study and practice guide built from four documents: product sheet, CD indication (US label), dosing and technique (US label) and AbbVie clinical letter (Charles 2012, Marsh 2014, Greene 1990), plus the team's anatomy slides.

MUST-KNOW SUMMARY

- › **Mechanism:** acetylcholine release inhibitor and neuromuscular blocking agent.
- › **CD indication:** adults only; goals: reduce **abnormal head position** and **neck pain**.
- › **Adult ceiling: 400 U / 3 months**, cumulative across all indications. **Pediatric:** the lower of **10 U/kg** and **340 U**.
- › **Typical CD dose:** mean **236 U** (P25–P75: 198–300 U) in previously treated patients. Toxin-naïve patients: start lower.
- › **SCM $\leq$ 100 U** to reduce dysphagia. **$\leq$ 50 U per site**, 25–30 G needle. Dilutions: 200 U/2 mL, 200 U/4 mL, 100 U/1 mL, 100 U/2 mL in preservative-free 0.9% NaCl.
- › **Boxed warning:** distant spread of toxin effect (dysphagia, dysphonia, dyspnea, ptosis, diplopia...) from hours to weeks; deaths have occurred. Reported at CD doses and lower.
- › **Reinject every ~12 weeks at the earliest.** Mean duration of effect $\approx$ 93–95 days; totals $\geq$ 180 U last ~3 weeks longer than $<$ 180 U.
- › **Start with primary muscles** for the posture; add secondary muscles only for a reason (Module 10).

Module 1 · The product and its indications

Botox is classified as an **acetylcholine release inhibitor** and a **neuromuscular blocking agent**. It has nine approved US indications, grouped here by system to make them easier to memorize:

Group	Indication	Population / condition
Urologic	Overactive bladder (urgency, frequency, urge incontinence)	Adults with inadequate response to or intolerance of anticholinergics
Urologic	Detrusor overactivity incontinence of neurologic cause (SCI, MS)	Adults with inadequate response to or intolerance of anticholinergics
Urologic	Neurogenic detrusor overactivity	Pediatric $\geq$ 5 years, same condition
Neurologic	Chronic migraine prophylaxis	Adults; $\geq$ 15 days/month with headache lasting $\geq$ 4 h
Neurologic	Spasticity	$\geq$ 2 years
Neurologic	Cervical dystonia	Adults; abnormal head position and neck pain
Neurologic	Blepharospasm associated with dystonia	$\geq$ 12 years
Ophthalmic	Strabismus	$\geq$ 12 years
Dermatologic	Severe axillary hyperhidrosis	Adults, inadequately managed with topical agents

Limitations of use (safety and efficacy not established): episodic migraine prophylaxis ($\leq$ 14 days/month) and hyperhidrosis in areas other than the axilla.

⚠ **Boxed warning: distant spread of toxin effect**

- › **Symptoms:** asthenia, generalized weakness, diplopia, ptosis, dysphagia, dysphonia, dysarthria, urinary incontinence and breathing difficulties.
- › **Onset:** hours to weeks after injection. Swallowing and breathing problems can be life-threatening; deaths have been reported.
- › Risk is probably greatest in children treated for spasticity, but it also occurs in adults, especially with predisposing conditions.
- › Reported in approved and unapproved uses, at doses comparable to those used in CD and spasticity, and lower.

Tips to remember Module 1

- › **Age rule:** 2 (spasticity), 5 (pediatric neurogenic detrusor), 12 (blepharospasm and strabismus), adults for everything else.
- › **Chronic migraine = 15/4:** ≥ 15 days per month, ≥ 4 hours per episode. Anything less is episodic and NOT approved.
- › The three urologic indications share the same condition: failure of or intolerance to anticholinergics.
- › When explaining the boxed warning to a CD patient, the key message is that the CD dose does not protect against the risk.

Module 2 · Dosing and injection technique in CD

General principles from the label

- Follow the indication-specific recommendations and use the **lowest recommended dose** when starting.
- **Adults:** maximum cumulative dose of **400 U in 3 months**, adding up all indications treated.
- **Pediatrics:** do not exceed the **lower of 10 U/kg and 340 U** in 3 months.
- Know the neuromuscular and structural anatomy, including changes from surgery or disease, especially when injecting **near the lungs**.
- Familiarity with standard **electromyographic techniques** can be useful.

Dose in cervical dystonia

The label relies on a double-blind study in patients who already tolerated Botox with an individualized dose: mean **236 U**, P25–P75 range **198 to 300 U**, divided among the affected muscles.

Adjust the dose at each session using five factors:

Factor	What to look at
1. Head and neck position	Dystonia pattern (torticollis, laterocollis, anterocollis, retrocollis, shift) → muscle selection
2. Location of pain	Painful muscles are usually the most active
3. Muscle hypertrophy	Sign of chronic overactivity; guides muscle and dose
4. Patient response	Effect and duration of the previous cycle
5. Adverse event history	Previous dysphagia or weakness means lowering the dose, especially in the SCM

Technique

- **Recommended dilutions:** 200 U/2 mL, 200 U/4 mL, 100 U/1 mL or 100 U/2 mL with **preservative-free 0.9% NaCl**, depending on volume and number of sites.
- **Maximum 50 U per site**, sterile **25–30 G** needle of appropriate length.
- **SCM ≤ 100 U** in total may decrease dysphagia.
- Localization with **EMG guidance** may be useful.

Tips to remember Module 2

- › Three numbers that get confused: **400 U** (label ceiling, 3 months), **360 U** (maximum per visit in Charles 2012), **340 U** (absolute pediatric ceiling). Only the first applies to adult practice.
- › Practical concentration: 100 U/1 mL = 10 U per 0.1 mL; 100 U/2 mL = 5 U per 0.1 mL. The second makes it easier to split small doses across many sites.
- › The SCM lies against the pharynx: high dose or spread there = dysphagia. That is why the label singles it out.
- › "Preservative-free" is not a minor detail: it is what the label requires for reconstitution.
- › For TWSTRS and the monitoring app: factors 1 and 2 (position and pain) are exactly the domains the scale quantifies.

Module 3 · Injection session protocol, step by step

Follow the steps in order. The source of each rule is in brackets; items without brackets are common clinical practice.

Step 1 · Before you start: check

Check	If not met
Adult with cervical dystonia (the indication is adults only) [label]	Do not inject under this indication.
≥ 12 weeks since the last session [label: ceiling per 3 months]	Reschedule. Do not shorten the interval.
Units from all indications in the last 3 months (migraine, bladder, etc.) [label]	Subtract what was already used from the 400 U ceiling.
No infection at the injection sites and no known hypersensitivity to the toxin [label]	Do not inject that day.
Pre-existing neuromuscular disease (myasthenia, Lambert-Eaton, ALS), baseline dysphagia or breathing problems [label]	Higher risk of systemic effects: lower dose and close monitoring.
Drugs that potentiate the block: aminoglycosides, curare-like agents, anticholinergics, muscle relaxants or another botulinum toxin [label]	Weigh the risk, adjust dose and warn the patient.

Step 2 · Ask about the previous cycle

- **Effect:** when did improvement start, when was it best, and when did the posture come back? Note the week of each.
- **Adverse events:** trouble swallowing, voice change, neck or shoulder weakness, trouble breathing.
- **What still bothers them:** the component or muscle that improved least (this drives the changes in Step 4).
- In CD-Track, review the **cycle curve** and the **pattern** at each timepoint before examining.

Step 3 · Examine and classify

- **Posture:** classify each component (rotation, tilt, flexion/extension, shift, shoulder) and its side from the patient's perspective (Module 6). Compare with the pattern reported by the app.
- **Palpate:** pain, hypertrophy and active contraction in each candidate muscle. Note which ones.
- **Score the TWSTRS** for the day, before injecting.

Step 4 · Choose muscles and dose

- **Muscles:** start with the **primary** muscles for the pattern (Module 8). Add secondary muscles only for a reason: pain, hypertrophy, an associated component, shoulder elevation, positive EMG or incomplete previous response (Module 10).
- **Do not use** muscles marked * (not approved for onabotA).

Patient type	Total dose to use
Toxin-naïve (first time)	Start low and adjust by response [label]. A conservative reference is the Greene templates: 140-165 U total depending on pattern.
Previously treated, good response	Repeat the dose and muscles that worked. Reference for treated patients: mean 236 U , usual 198-300 U [label].
Previously treated, short effect	Increase the total dose within the ceilings. Totals ≥ 180 U are associated with ~3 more weeks of effect [Marsh 2014].
Previous dysphagia or weakness	Lower the dose in the responsible muscle (especially the SCM) and redistribute [label].

Limits you do not cross

- › **400 U** cumulative in 3 months, adding up all indications.
- › **≤ 50 U per site.**
- › **SCM ≤ 100 U** in total. When possible, avoid injecting both SCMs in the same session.
- › Change **one variable per cycle** (one muscle or one dose), so you know what caused the change.

Step 5 · Prepare

Vial	Diluent (preservative-free 0.9% NaCl)	Concentration	Use it when
100 U	1 mL	10 U / 0.1 mL	Few sites, high dose per site
100 U	2 mL	5 U / 0.1 mL	Many small sites (easier to split)
200 U	2 mL	10 U / 0.1 mL	Few sites, high total dose
200 U	4 mL	5 U / 0.1 mL	Many sites, high total dose

- Sterile **25–30 G** needle, of a length suited to the muscle [label].
- Use the reconstituted vial within **24 hours** and keep it refrigerated (2–8 °C) in the meantime [label].

Step 6 · Inject

- Split each muscle's dose into several equal sites along the muscle belly.
- Use **EMG or ultrasound guidance** for deep or high-risk muscles: scalenes, longissimus, levator scapulae [label: EMG may be useful].
- In the SCM, inject the upper and middle thirds and avoid the deep area near the pharynx.

Step 7 · Before the patient leaves

- **Give the warning signs** (Module 5): trouble swallowing, speaking or breathing, within hours or weeks → go to the emergency department immediately.
- **Tell them what to expect:** improvement within the first 2 weeks, peak around week 6, gradual return around weeks 10–12.
- **Record** muscle, side, units, sites, dilution, lot, guidance used and adverse events (template in Module 5).
- In **CD-Track**, enter the injection date as **baseline**. The app builds the follow-up calendar.

Worked example: right torticollis, toxin-naïve patient

- › **Pattern:** the chin turns right → contralateral = left, ipsilateral = right.
- › **Starting template** (Greene pure rotational): **left** SCM 40 U (8 sites) + **right** trapezius 55 U (11 sites) + **left** trapezius 45 U (9 sites) = **140 U**.
- › **Adjust with the anatomy slide:** consider adding the **right splenius** (ipsilateral primary) if it is active or painful.
- › **Preparation:** 100 U in 2 mL (5 U/0.1 mL) → each 5 U site = **0.1 mL**. 140 U = 2.8 mL: you need 2 × 100 U vials, or 1 × 200 U vial in 4 mL.
- › **Check:** 140 U ≤ 400 · 5 U per site ≤ 50 · SCM 40 U ≤ 100 ✓
- › *Educational example, not a prescription: individualize with the label's five factors (Module 2).*

Module 4 · Following the cycle and deciding on reinjection

What to do at each CD-Track timepoint

Timepoint	What to expect	What to do
Baseline (injection day)	Posture without toxin effect	Record before injecting. Full TWSTRS. This is the reference for the whole cycle.
1 week	No or early improvement	Check adverse events (swallowing, voice, weakness). Do not judge efficacy yet.
1 month	Near peak effect (weeks 2-6)	Judge the muscle selection: which component improved and which did not? Note it for the next session.
2 months	Sustained effect	If the curve is already returning toward baseline → short effect; note it to adjust the dose.
3 months (pre-injection)	Effect wearing off (weeks 10-12)	Decide the next session with the table below. This recording becomes the next cycle's baseline.

Decision table at the reinjection visit

If you see...	Do this
Good response lasting to ~12 weeks, no adverse events	Repeat the same scheme (same muscles, sides and doses).
Good response but lost before ~10 weeks	Do not shorten the interval. Increase the total dose within the ceilings (≥ 180 U is associated with longer duration) or reinforce the muscle of the component that relapses first.
One component improves and another persists (e.g., residual tilt)	Keep what worked and add the muscles for the residual component (Module 8).
Posture improves but pain persists	Look for the painful or hypertrophic muscle that was not injected and include it.
Poor response in the first cycle	Review the pattern and muscle selection, use EMG/ultrasound guidance and increase the dose gradually.
Used to respond well and stops responding after several cycles, with correct technique and dose	Suspect neutralizing antibodies : refer or review with the team. Always use the lowest effective dose at the longest feasible interval [label].
Dysphagia	Lower the SCM (≤ 100 U, ideally less), avoid both SCMs in the same session and redistribute toward posterior muscles.
Neck weakness or difficulty holding the head up	Reduce the dose in the posterior extensors , especially if injected bilaterally.
Shoulder weakness or drop	Reduce the trapezius dose.
CD-Track changes pattern between cycles (e.g., torticollis to mixed)	Re-examine: the new component needs its own muscles.

Reinjection rules

- › Reinject every **~12 weeks** at the earliest. Mean duration of effect is ~13 weeks (93-95 days) [Marsh 2014].
- › If the effect does not reach 12 weeks, adjust the **dose or the muscles**, not the interval.
- › Change **one thing per cycle** and compare the CD-Track curve with the previous cycle's.
- › The CD-Track "next injection due" date is the 3-month visit.
- › **Golden rules:** 400 U in 3 months · 50 U per site · 100 U SCM · 12 weeks between sessions · 180 U for longer duration. Primary muscles first; secondary only for a reason; warning signs explained and session documented.

Module 5 · Safety and special situations

What to tell the patient (warning signs)

Patient script

- › "The effect of the toxin can spread beyond the injection site. This can happen from a few hours up to several weeks later."
- › "If you have **trouble swallowing, speaking or breathing**, go to the emergency department right away."
- › "Also let me know if you notice general weakness, drooping eyelids, double vision, a hoarse voice or neck weakness."
- › "It is normal for improvement to take up to two weeks to start."

Remember: the risk also exists at the doses used for cervical dystonia, and at lower doses [label, boxed warning].

Special situations

Situation	What to do
Patient coming from another botulinum toxin	Do not convert units 1:1: onabotA units are not interchangeable with other products [AbbVie]. Start as a toxin-naïve patient.
Receives Botox for another indication (migraine, bladder, spasticity)	Add up all units in the quarter: the 400 U ceiling is shared [label].
Isolated anterocollis	No reference dosing template. Deep muscles near the pharynx: use imaging guidance, a cautious dose, or refer to an experienced injector.
Muscles marked * on the slides	Do not include them in an onabotA plan: they are not approved.
Dysphagia or weakness in the previous cycle	Lower the dose of the responsible muscle and document it before the next session.
Patient with neuromuscular disease	Lower dose, close monitoring and an explicit discussion of the risk [label].

What to record at every session

Field	Example
Date, lot and vial(s)	2026-10-01 · lot ____ · 1 × 200 U
Dilution	200 U in 4 mL (5 U/0.1 mL)
Clinical pattern and CD-Track pattern	Right torticollis · CD-Track: right torticollis (yaw 32°)
Per muscle: side, units and sites	L SCM 40 U / 8 · R trapezius 55 U / 11 · R splenius 30 U / 3
Guidance used	EMG for splenius
Session total and quarter total	125 U · 125 U (no other indications)
Adverse events from the previous cycle	None
TWSTRS for the day	Severity __ / 35

Part 2 · Applied anatomy: posture types and muscles

Uses the anatomy slides from the CD-Track Learning Hub: how each posture type is defined, how the app classifies it from the videos, and which muscles go with each pattern.

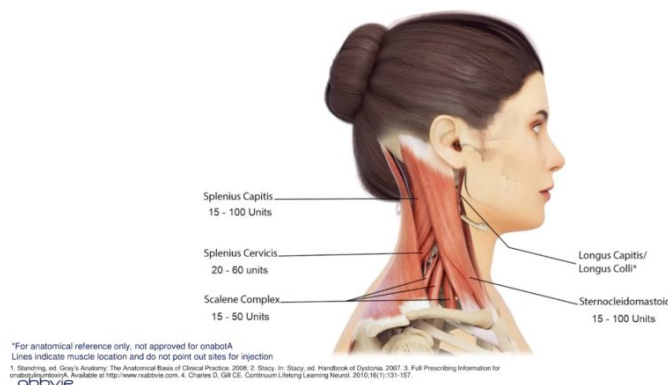
MUST-KNOW (Part 2)

- › **The type is defined by the plane of movement:** rotation (torticollis) = vertical axis / yaw; tilt (laterocollis) = coronal plane / roll; flexion and extension (antero/retrocollis) = sagittal plane / pitch.
- › **The side is named by the direction of the head:** right torticollis = the chin points to the patient's right; right laterocollis = the right ear moves toward the right shoulder.
- › **Ipsi- and contralateral are relative to that direction**, not to the observer. Right torticollis → **left** SCM (contralateral) + **right** splenius (ipsilateral).
- › **Laterocollis is all ipsilateral:** scalenes, SCM, splenius, levator and trapezius on the side of the tilt. Antero- and retrocollis are **bilateral**.
- › **CD-Track** counts an axis as "active" if its mean resting angle is $\geq 10^\circ$. If the largest is more than $1.5\times$ the second, that is the pattern; otherwise it is **mixed**.
- › **Primary** (bold blue on the slides) = starting point; **secondary** (regular blue) = added based on exam or response; * = anatomical reference only, **not approved** for onabotA.
- › **Per-muscle ranges** (slides): SCM 15-100 U · splenius capitis 15-100 · splenius cervicis 20-60 · trapezius 20-100 · levator 20-100 · scalenes 15-50 · longissimus 30-100 · semispinalis capitis 30-100.

Part 2 contents

- **Module 6** · How posture types are defined and how CD-Track classifies them
- **Module 7** · What classifying the type is for (in clinic and in the app)
- **Module 8** · Posture atlas: torticollis, laterocollis, antero/retrocollis and combined posture
- **Module 9** · Muscle atlas: action, attachments, dose range and postures involved
- **Module 10** · Primary vs secondary muscles, and self-assessment

*Main muscles involved in Cervical Dystonia



Overview of the main muscles in cervical dystonia with per-muscle onabotA ranges. Longus capitis/colli (*) is anatomical reference only, not approved for onabotA.

Module 6 · How posture types are defined

Cervical dystonia is described by **which way the head deviates** at rest. Each type corresponds to a plane of movement and, in CD-Track, to one of the three head-pose angles computed by computer vision (yaw, roll and pitch). The clinical definition and the app's measurement therefore line up: the axis that defines the type is the one plotted across the cycle.

Type	Movement	Plane / axis	How the side is named	CD-Track measure	TWSTRS item
Torticollis (rotation)	The chin turns toward a shoulder	Vertical axis · yaw	By the side the chin points to	Mean yaw, frontal resting video	A1 (0-4)
Laterocollis (tilt)	The ear moves toward the shoulder	Coronal · roll	By the side it tilts toward	Mean roll, frontal resting video	A2 (0-3)
Anterocollis (flexion)	The chin drops toward the chest	Sagittal · pitch	No side: bilateral	Mean pitch of R and L profiles (averaged)	A3a (0-3)
Retrocollis (extension)	The head goes back	Sagittal · pitch	No side: bilateral	Mean pitch of the profiles (opposite sign)	A3b (0-3)
Lateral / sagittal shift	The head translates without turning	Translation	By side, or forward/back	Head offset vs shoulder midline; ear vs shoulder	A4, A5 (0-1)
Shoulder elevation	One shoulder higher than the other	Shoulder girdle	By the elevated shoulder	Ear-shoulder asymmetry in the shoulder check	D (0-3)
Combined / mixed	Two or more components at once	Several	Each component with its side	See classification rule below	Sum of items

Side convention: the most confusing part

- › **The side of the posture is always the patient's** (anatomical right), never the observer's or the screen's. The front camera image can be mirrored; the app converts the sign of the angle to the real side.
- › **Ipsilateral / contralateral are defined relative to the direction of each component:** for torticollis, relative to the side the chin turns to; for laterocollis, to the side of the tilt; for shoulder elevation, to the elevated shoulder.
- › **Worked example:** right torticollis → left SCM, left upper trapezius and left anterior scalene (contralateral) + right splenius, right levator and right longissimus (ipsilateral).
- › The same muscle can be "contralateral" for one component and "ipsilateral" for another (see the combined case in Module 8).

CD-Track classification rule (as implemented)

- **1. Measure.** At each timepoint, take the mean angles from the resting, eyes-closed videos: yaw and roll from the frontal view, and pitch as the average of the right and left profiles.
- **2. Filter.** An axis counts as a component only if its mean angle is $\geq 10^\circ$. Below that it is treated as measurement noise and normal variation.
- **3. No components.** If no axis reaches 10° , the pattern is **indeterminate** (head near neutral, or marked improvement after toxin).
- **4. One component.** That is the pattern, with its direction: right/left torticollis, right/left laterocollis, anterocollis or retrocollis.
- **5. Two or more.** They are ranked by magnitude. If the largest is ≥ 1.5 times the second, it dominates and sets the pattern; otherwise the pattern is **mixed**. The angle reported as the main measure is always that of the largest component.

Example (mean angles)	Calculation	Classification
Yaw 32° to the right, roll 6°, pitch 4°	Only yaw $\geq 10^\circ$	Right torticollis (32°)
Yaw 32°, roll 14° to the left	$32 / 14 = 2.3 \rightarrow \geq 1.5$	Right torticollis (32°)
Yaw 22°, roll 18°	$22 / 18 = 1.2 \rightarrow < 1.5$	Mixed (22°)
Pitch 16° backward, yaw 8°	Only pitch $\geq 10^\circ$	Retrocollis (16°)
Yaw 7°, roll 5°, pitch 6°	None $\geq 10^\circ$	Indeterminate

What is not part of the classification, and limits

- › **Shifts** (A4, A5) and **shoulder elevation** (D) are measured and scored in the TWSTRS but **do not change the pattern**: the pattern is decided only from yaw, roll and pitch. In clinic they do change muscle selection.
- › The **10°** and **1.5x** thresholds are **provisional** and pending calibration with the team.
- › **Pitch** (antero/retrocollis) is the noisiest measure: it comes from the profiles, where the face mesh degrades.
- › Absolute angles carry perspective bias; what is reliable is **within-patient change**.
- › **Collis vs caput**: the classic nomenclature does not distinguish whether movement occurs between head and neck (C1-C2, "-caput") or between neck and trunk ("-collis"). The Col-Cap concept separates the two levels because the muscles involved differ. CD-Track measures total head orientation relative to the camera and shoulders, so it **adds both levels together**. That distinction is left to the clinical exam.

Module 7 · What classifying the type is for

Classifying the posture is not an end in itself. It is the step that connects what you see (or measure on video) with the decision on which muscles to treat and how to follow the response. Its concrete uses:

Use	In clinic	In CD-Track
1. Choose muscles	The type is label factor 1 for selecting muscles (Module 2). Each posture has its list of primary and secondary muscles.	The pattern opens the matching posture slide in the Learning Hub and in the patient's injection plan.
2. Resolve the real side	Translates "contralateral/ipsilateral" into the patient's "right/left".	The detected direction (e.g., chin to the right) becomes the Muscle / Side / Dose table.
3. Starting dose template	Greene 1990 gives a fixed scheme per pattern. Use it as a template, not a recipe.	Torticollis → pure rotational · laterocollis → tilt · retrocollis → torticollis + retrocollis · mixed → tilt (the broadest) · anterocollis → no template.
4. Choose what to measure	The TWSTRS scores each component separately (A1–A5).	The dominant-axis angle is the main measure plotted across the cycle (baseline → 3 months).
5. Detect incomplete response	Rotation that improves but leaves a residual tilt → review the muscles of the remaining component.	A pattern change between timepoints (e.g., from torticollis to mixed or laterocollis) signals that muscle selection should be reviewed at the next session.
6. Common language	Raters, residents and coordinators describe the posture the same way.	Allows the app's classification to be compared with the blinded rater's.

What the classification does NOT do

- › It does not diagnose dystonia or distinguish it from other causes of abnormal head posture (orthopedic, ocular, functional, head drop from weakness).
- › **It does not prescribe.** The dose is individualized with the label's five factors: position, pain, hypertrophy, previous response and adverse events.
- › It does not replace palpation or EMG/ultrasound for localizing muscles, especially deep ones.
- › It does not separate caput from collis (Module 6), and it cannot see contractions of muscles that do not visibly move the head.

Tips to remember Modules 6 and 7

- › Three axes, three types: **yaw = shaking "no"** (torticollis), **roll = ear to shoulder** (laterocollis), **pitch = nodding "yes"** (antero/retrocollis).
- › To avoid getting the side wrong, always think from the patient: "which way does THEIR chin point?"
- › The SCM is the exception to memorize: **contralateral** in torticollis but **ipsilateral** in laterocollis.
- › "Mixed" in the app does not mean "unusual": it means no component dominates by 1.5×. It is the most common presentation in clinic.

Module 8 · Posture atlas

Each slide lists the muscles that produce the posture. **Bold** = primary contributor; regular = secondary; * = anatomical reference only, not approved for onabotA.

8.1 · Torticollis (rotation)

Single posture:

Torticollis (Rotation)

Contralateral

Sternocleidomastoid

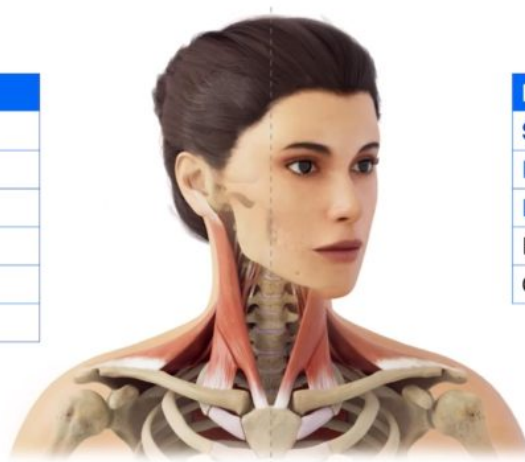
Trapezius (upper)

Anterior scalene

Semispinalis cervicis*

Multifidus*

Rotatores muscles*



Ipsilateral

Splenius capitis/cervicis

Levator scapulae

Longissimus (capitis/cervicis)

Rectus capitis posterior major*

Obliquus capitis inferior*

Bold blue primary contributor, standard blue secondary contributor

*For anatomical reference only, not approved for onabotA

1. Standing, ed. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008; 2. Stacy, In: Stacy, ed. Handbook of Dystonia. 2007; 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.rxabbvie.com>; 4. Charles D. Gill CE. Continuum Lifelong Learning Neurol. 2010;16(1):131-157.

abbvie

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Team anatomy slide (AbbVie). Lines show muscle location, not injection sites.

Side	Primary	Secondary	* Not approved for onabotA
Contralateral to chin rotation	Sternocleidomastoid	Upper trapezius · Anterior scalene	Semispinalis cervicis · Multifidus · Rotatores
Ipsilateral to chin rotation	Splenius capitis / cervicis	Levator scapulae · Longissimus (capitis/cervicis)	Rectus capitis posterior major · Obliquus capitis inferior

Clinical and app key points

- › **How CD-Track detects it:** mean yaw $\geq 10^\circ$ in the frontal resting video, direction from the sign. Scored in TWSTRS A1 (0 = none, 1 = 1-22°, 2 = 23-45°, 3 = 46-67°, 4 = 68-90°).
- › Rotation is work from **both sides**: contralateral SCM and ipsilateral splenius are the core pair.
- › **Difference from Greene 1990:** the template the app uses for pure rotation injects contralateral SCM and trapezius on both sides, **without splenius**. The slide lists the ipsilateral splenius as primary. Consider it even though the template leaves it out.

8.2 · Laterocollis (tilt)

Single posture:

Laterocollis (Tilt)

Ipsilateral

Anterior/middle/posterior scalenes

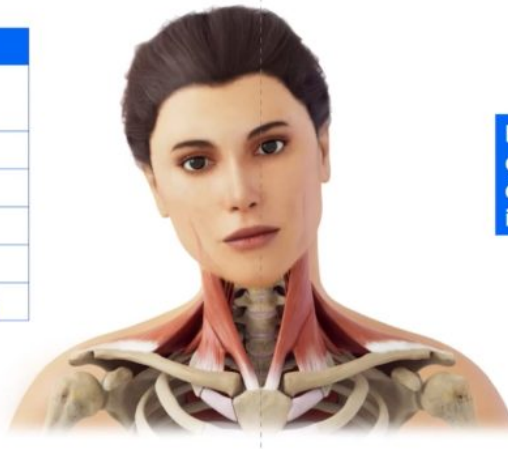
Sternocleidomastoid

Splenius capitis/cervicis

Levator scapulae

Trapezius (upper)

Longissimus (capitis/cervicis)



Note the presence/absence of shoulder elevation to determine full range of involved muscles

Bold blue primary contributor, standard blue secondary contributor.

1. Standing, ed. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008; 2. Stacy, In: Stacy, ed. Handbook of Dystonia. 2007; 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.nabbvie.com>; 4. Charles D. Gil CE. Continuum Lifelong Learning Neurol. 2010;16(1):131-157.

abbvie

4

Team anatomy slide (AbbVie). Lines show muscle location, not injection sites.

Side	Primary	Secondary
Ipsilateral to the tilt (all)	Anterior / middle / posterior scalenes · Sternocleidomastoid · Splenius capitis / cervicis · Levator scapulae · Upper trapezius	Longissimus (capitis/cervicis)

Clinical and app key points

- › **How CD-Track detects it:** mean roll $\geq 10^\circ$ in the frontal resting video. Scored in A2 (0 = none, 1 = 1-15°, 2 = 16-35°, 3 = > 35°). The TWSTRS scores the tilt **excluding** shoulder elevation; that is scored separately in D with the shoulder check.
- › All the work is on **one side**: the side of the tilt. Here the SCM is **ipsilateral**.
- › The slide asks you to check for **shoulder elevation**. If present, levator scapulae and trapezius matter more. That is item D, which the app already measures.
- › With five primaries on the same side, the dose budget is spread over many muscles: watch the total and the SCM ≤ 100 U limit.

8.3 · Anterocollis (flexion)

Single posture: Anterocollis (Flexion)

Bilateral
Anterior/middle scalenes
Longus capitis/colli*
Rectus capitis anterior*



Bold blue primary contributor
*For anatomical reference only, not approved for onabotA
1. Standing, ed. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008; 2. Stacy, In: Stacy, ed. Handbook of Dystonia. 2007. 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.rxabbvie.com>. 4. Charles D, Gill CE. Continuum Lifelong Learning Neurol. 2010;16(1):131-157.
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Team anatomy slide (AbbVie). Lines show muscle location, not injection sites.

Side	Primary	* Not approved for onabotA
Bilateral	Anterior / middle scalenes	Longus capitis / longus colli · Rectus capitis anterior

Clinical and app key points

- › **How CD-Track detects it:** mean forward pitch $\geq 10^\circ$, averaging the right and left profiles. Scored in A3a (0-3). This is the app's noisiest measure.
- › This is the posture **least covered by the evidence** in this guide: Charles 2012 excluded isolated anterocollis and Greene 1990 has no template. That is why the app does not suggest doses for this pattern.
- › The deep prevertebral flexors (*) lie next to the pharynx and are not approved: higher risk of **dysphagia**.
- › Distinguish anterocollis (chin drops, flexion) from anterior **sagittal shift** (the head moves forward without flexing, A5). They look alike in profile and involve different muscles.

8.4 · Retrocollis (extension)

Single posture:

Retrocollis (Extension)

Bilateral
Semispinalis capitis
Splenius capitis
Splenius cervicis
Longissimus (capitis/cervicis)
Trapezius (upper)
Levator scapulae
Semispinalis cervicis*
Rectus capitis posterior major/minor*
Obliquus capitis superior*



Bold blue primary contributor, standard blue secondary contributor

*For anatomical reference only, not approved for onabotA

1. Standring, ed. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008; 2. Stacy, In: Stacy, ed. Handbook of Dystonia. 2007; 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.rxabbvie.com>. 4. Charles D. Gill CE. Continuum Lifelong Learning Neurol. 2010;16(1):131-157.

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Team anatomy slide (AbbVie). Lines show muscle location, not injection sites.

Side	Primary	Secondary	* Not approved for onabotA
Bilateral	Semispinalis capitis · Splenius capitis · Splenius cervicis · Longissimus (capitis/cervicis)	Upper trapezius · Levator scapulae	Semispinalis cervicis · Rectus capitis posterior major/minor · Obliquus capitis superior

Clinical and app key points

- › **How CD-Track detects it:** mean backward pitch $\geq 10^\circ$ in the profiles. Scored in A3b (0–3). Only A3a or A3b counts, whichever applies.
- › All muscles are **posterior and bilateral**. Semispinalis capitis is the pure extensor on the list: it appears only in retrocollis and torticollis + retrocollis.
- › Treating extensors on both sides can cause **neck weakness** (reported in Greene 1990). Start conservatively.
- › App template: Greene torticollis + retrocollis (contralateral SCM and trapezius on both sides). It is designed for retrocollis **with** rotation. For pure retrocollis, the slide is the better guide.

8.5 · Combined posture: right torticollis + left laterocollis + left shoulder elevation

Combination posture:
Right torticollis, left laterocollis, left shoulder elevation



Right torticollis

Contralateral
Sternocleidomastoid
Trapezius (upper)
Anterior scalene

Ipsilateral
Splenius capitis
Splenius cervicis
Levator scapulae

Left laterocollis

Ipsilateral
Splenius capitis
Splenius cervicis
Trapezius (upper)
Anterior/middle/posterior scalenes

Left shoulder elevation

Ipsilateral
Levator scapulae
Trapezius

1. Standing, ed. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008; 2. Stacy, In: Stacy, ed. Handbook of Dystonia. 2007; 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.nabbvie.com>; 4. Charles D. Gill CE. Continuum Lifelong Learning Neurol. 2010;16(1):131-157.

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Team anatomy slide (AbbVie). Lines show muscle location, not injection sites.

Component	Side	Muscles (patient's real side)
Right torticollis	Contralateral = left	SCM · Upper trapezius · Anterior scalene
	Ipsilateral = right	Splenius capitis · Splenius cervicis · Levator scapulae
Left laterocollis	Ipsilateral = left	Splenius capitis · Splenius cervicis · Upper trapezius · Anterior/middle/posterior scalenes
Left shoulder elevation	Ipsilateral = left	Levator scapulae · Trapezius

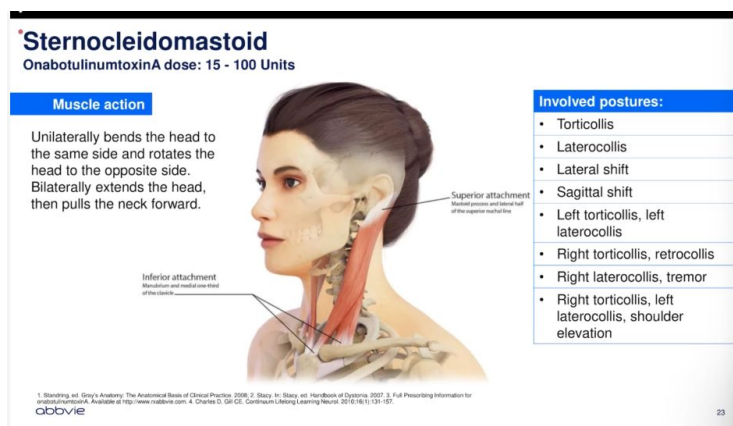
Clinical and app key points

- › **How CD-Track detects it:** yaw and roll $\geq 10^\circ$ with neither more than $1.5\times$ the other → **mixed** pattern. Shoulder elevation is scored in D. In this case the app uses the Greene tilt template as the broadest one, and the final selection is clinical.
- › **Method:** break the posture into components, look up each list and translate it to the real side.
- › The **left trapezius** appears in all three components. It is a high-yield muscle in this patient.
- › **Both splenii** appear: the right for the rotation and the left for the tilt. Here the exam decides which one is actually active.
- › The left SCM is contralateral to the right rotation. Although the SCM also tilts toward its own side, this slide does not list it for the left laterocollis.

Module 9 · Muscle atlas

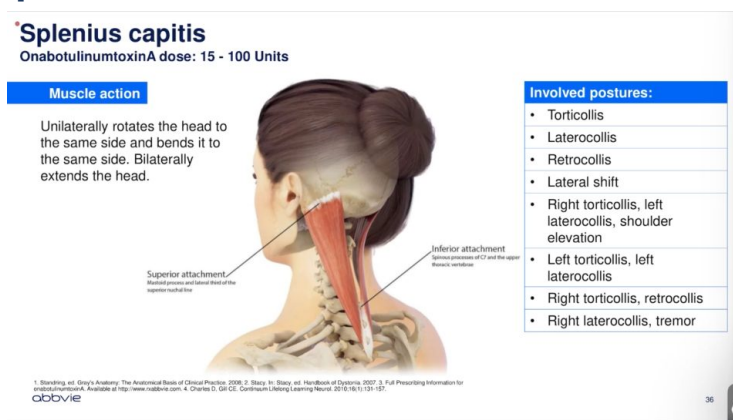
For each muscle: action, attachments, onabotA range shown on the slides (total dose per muscle per session, split into sites of ≤ 50 U) and the postures it is involved in. The "Postures" row combines the muscle and posture slides.

9.1 · Sternocleidomastoid (SCM)



Action	Unilateral: tilts the head to the same side and rotates it to the opposite side . Bilateral: extends the head, then pulls the neck forward.
Attachments	Superior: mastoid process and lateral half of the superior nuchal line. Inferior: manubrium and medial third of the clavicle.
onabotA dose	15-100 U (label: SCM $\leq$ 100 U to reduce dysphagia)
Postures	Torticollis (contralateral, primary) · Laterocollis (ipsilateral, primary) · Lateral and sagittal shift · combinations with retrocollis, laterocollis or tremor.
Practical note	The muscle most tied to dysphagia , because of its closeness to the pharynx. The classic contralateral muscle.

9.2 · Splenius capitis



Action	Unilateral: rotates the head to the same side and tilts it to the same side. Bilateral: extends the head.
Attachments	Superior: mastoid process and lateral third of the superior nuchal line. Inferior: spinous processes of C7 and upper thoracic vertebrae.
onabotA dose	15-100 U
Postures	Torticollis (ipsilateral, primary) · Laterocollis (primary) · Retrocollis (bilateral, primary) · Lateral shift.
Practical note	The ipsilateral "partner" of the contralateral SCM in rotation. Superficial and palpable under the trapezius.

9.3 · Splenius cervicis

Splenius cervicis
OnabotulinumtoxinA dose: 20 - 60 Units

Muscle action

Unilaterally rotates the upper neck and bends it to the same side. Bilaterally extends the upper cervical spine.

Superior attachment
Transverse process of the upper cervical vertebrae (C1 to C3)

Inferior attachment
Spinous processes of the middle thoracic vertebrae (T3 to T6)

Involved postures:

- Torticollis
- Laterocollis
- Retrocollis
- Lateral shift
- Right torticollis, left laterocollis, shoulder elevation
- Left torticollis, left laterocollis
- Right torticollis, retrocollis
- Right laterocollis, tremor

1. Standring et al. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008. © Stanley H. Standring et al. Handbook of Dyspraxia. 2007. © Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.mrbd.com>. 4. Charles D. Gill CE. Continuous Learning Learning Network. 2016;16(1):131-137.

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Action	Unilateral: rotates the upper neck and tilts it to the same side. Bilateral: extends the upper cervical spine.
Attachments	Superior: transverse processes of C1-C3. Inferior: spinous processes of T3-T6.
onabotA dose	20-60 U
Postures	Torticollis · Laterocollis · Retrocollis · Lateral shift · combinations.
Practical note	Together with the scalenes, the lowest range on the list. Deeper and more caudal than splenius capitis.

9.4 · Trapezius (upper)

Trapezius (upper)
OnabotulinumtoxinA dose: 20 - 100 Units

Muscle action

Unilaterally rotates the head to the opposite side and bends it to the same side. Bilaterally extends the head.

Superior attachment
Medial one-third of the superior nuchal line and the external occipital protuberance

Inferior attachment
Lateral one-third of the clavicle, acromion process, and scapular spine; spinous processes of the thoracic and lower cervical vertebrae

Involved postures:

- Torticollis
- Laterocollis
- Retrocollis
- Lateral shift
- Right torticollis, left laterocollis, shoulder elevation
- Left torticollis, left laterocollis
- Right torticollis, retrocollis
- Right laterocollis, tremor

1. Standring et al. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008. © Stanley H. Standring et al. Handbook of Dyspraxia. 2007. © Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.mrbd.com>. 4. Charles D. Gill CE. Continuous Learning Learning Network. 2016;16(1):131-137.

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Action	Unilateral: rotates the head to the opposite side and tilts it to the same side. Bilateral: extends the head.
Attachments	Superior: medial third of the superior nuchal line and external occipital protuberance. Inferior: lateral third of the clavicle, acromion and scapular spine; spinous processes of the thoracic and lower cervical vertebrae.
onabotA dose	20-100 U
Postures	Torticollis (contralateral, secondary) · Laterocollis (ipsilateral, primary) · Retrocollis (secondary) · Lateral shift · shoulder elevation .
Practical note	Easy to inject, so it is often overused. Excess weakening can cause shoulder drop or difficulty raising the arm.

9.5 · Levator scapulae

Levator scapulae

OnabotulinumtoxinA dose: 20 - 100 Units

Muscle action

Unilaterally rotates the neck to the same side and bends it to the same side. Bilaterally extends the neck.



Involved postures:

- Torticollis
- Laterocollis
- Retrocollis
- Lateral shift

- Right torticollis, left laterocollis, shoulder elevation
- Left torticollis, left laterocollis

1. Stanning, ed. Gray's Anatomy: The Anatomical Basis of Clinical Practice. 2008. 5. Stacy, ed. Handbook of Oculoplasty. 2007. 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.novartis.com>. 4. Chertin G, Gil CE. Continuous levator scapulae spasm. 2015;6:1461-139-137

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Action	Unilateral: rotates the neck to the same side and tilts it to the same side. Bilateral: extends the neck.
Attachments	Superior: transverse processes of C1–C4. Inferior: superior medial border of the scapula.
onabotA dose	20–100 U
Postures	Torticollis (ipsilateral, secondary) · Laterocollis (primary) · Retrocollis (secondary) · Lateral shift · shoulder elevation.
Practical note	The muscle that links to TWSTRS item D. If the shoulder check shows asymmetry, it is a candidate.

9.6 · Scalene complex

*Scalene complex

OnabotulinumtoxinA dose: 15 - 50 Units

Muscle action

Anterior scalene = Unilaterally bends the neck to the same side and rotates the head to the opposite side.
Bilaterally flexes the neck

Middle scalene = Unilaterally bends the neck to the same side. Bilaterally flexes the neck

Posterior scalene = Unilaterally bends the neck to the same side

Superior attachment
Anterior scalene - Transverse process of the middle cervical vertebrae (C2 to C6)
Middle scalene - First rib posterior to the vertical vertebral line (C2 to C6)
Posterior scalene - Transverse processes of the middle cervical vertebrae (C4 to C6)

Inferior attachment
Anterior scalene - First rib anterior to the groove for the subclavian artery
Middle scalene - First rib posterior to the groove for the subclavian artery
Posterior scalene - Posterior aspect of the second rib

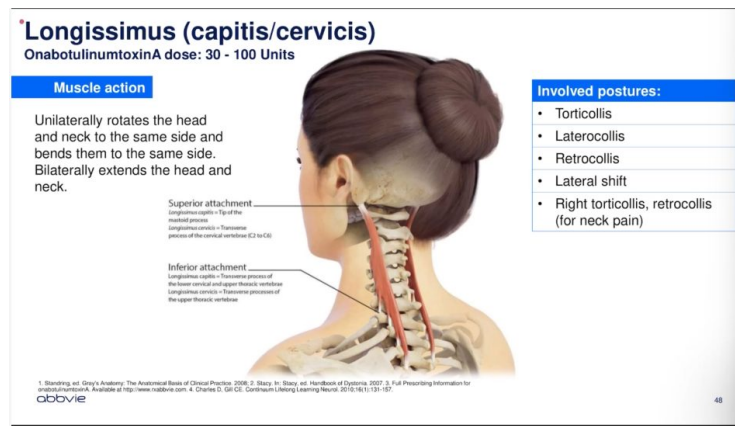
1. Shandling, et. Gray's Anatomy: The Anatomical Basis of Clinical Practice, 2009; 2. Slacy, et. Handbook of Dyslexia, 2007; 3. Full Prescribing Information for onabotulinumtoxinA. Available at <http://www.novartis.com>; 4. Chavakis G, Gell-Clayton, Learning Nursing, 2010; 16(1): 131-137.

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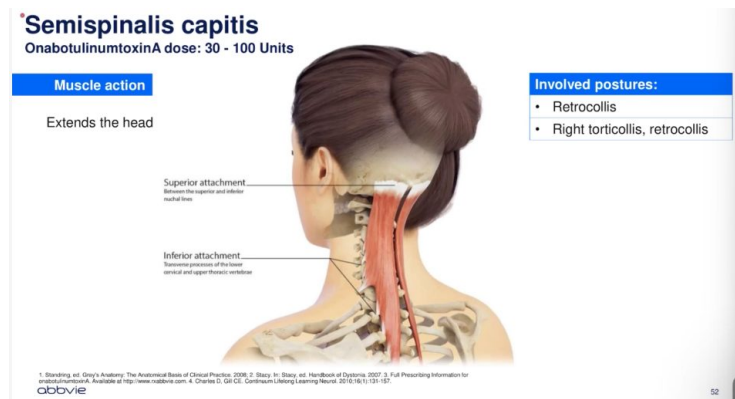
Action	Anterior: tilts the neck to the same side and rotates the head to the opposite side; bilaterally flexes the neck. Middle: tilts to the same side; bilaterally flexes. Posterior: tilts to the same side.
Attachments	Superior: transverse processes of C3–C6 (anterior), C2–C7 (middle) and C4–C6 (posterior). Inferior: first rib in front of (anterior) and behind (middle) the groove for the subclavian artery; posterior aspect of the second rib (posterior).
onabotA dose	15-50 U
Postures	Torticollis (anterior, contralateral, secondary) · Laterocollis (all three, primary) · Anterocollis (anterior/middle, bilateral, primary).
Practical note	Lies next to the brachial plexus and subclavian artery , and close to the pharynx. Best injected under EMG or ultrasound guidance.

9.7 · Longissimus (capitis / cervicis)



Action	Unilateral: rotates the head and neck to the same side and tilts them to the same side. Bilateral: extends the head and neck.
Attachments	Superior: tip of the mastoid (capitis); transverse processes of C2-C6 (cervicis). Inferior: transverse processes of the lower cervical and upper thoracic vertebrae (capitis); transverse processes of the upper thoracic vertebrae (cervicis).
onabotA dose	30-100 U
Postures	Torticollis (ipsilateral, secondary) · Laterocollis (secondary) · Retrocollis (primary) · Lateral shift · right torticollis + retrocollis (for neck pain).
Practical note	Deep and hard to localize by palpation. A candidate for guided injection.

9.8 · Semispinalis capitis



Action	Extends the head.
Attachments	Superior: between the superior and inferior nuchal lines. Inferior: transverse processes of the lower cervical and upper thoracic vertebrae.
onabotA dose	30-100 U
Postures	Retrocollis (bilateral, primary) · right torticollis + retrocollis.
Practical note	The pure extensor on the list. If it appears in the plan, the pattern has an extension component. Do not confuse it with semispinalis cervicis (*), which is not approved.

Muscle	Range (U)	Torticollis	Laterocollis	Anterocollis	Retrocollis
SCM	15-100	Contra · P	Ipsi · P	—	—
Splenius capitis/cervicis	15-100 / 20-60	Ipsi · P	Ipsi · P	—	Bilat · P
Upper trapezius	20-100	Contra · S	Ipsi · P	—	Bilat · S
Levator scapulae	20-100	Ipsi · S	Ipsi · P	—	Bilat · S
Scalenes	15-50	Anterior contra · S	Ipsi · P	Ant/middle bilat · P	—
Longissimus	30-100	Ipsi · S	Ipsi · S	—	Bilat · P
Semispinalis capitis	30-100	—	—	—	Bilat · P

Summary table: P = primary, S = secondary; "contra" and "ipsi" relative to the direction of each component. Built from the posture slides.

Module 10 · Primary vs secondary muscles, and self-assessment

When to add a secondary muscle

The practical rule: **start with the primary muscles and add secondary ones when something justifies it**. In a toxin-naïve patient with a simple posture, primaries only at a conservative dose tells you what worked and lowers the risk of weakness or dysphagia.

Situation	What it suggests	What CD-Track adds
Incomplete response to the previous cycle	Another muscle is contributing; find it before raising the dose in the primaries	The dominant-axis curve drops little or returns earlier than expected
Hypertrophy, pain or palpable contraction in a secondary muscle	Include it: the label uses pain and hypertrophy to choose muscles	— (clinical exam)
Shoulder elevation	Levator scapulae and upper trapezius	Item D from the shoulder check
Associated component (e.g., tilt in a torticollis)	Add the muscles for that component	Roll or pitch $\geq 10^\circ$; mixed pattern
EMG or ultrasound showing dystonic activity	Include it, especially if deep (longissimus, scalenes)	—
Previous dysphagia or weakness	Lower the SCM and redistribute toward posterior and secondary muscles	— (history)

Safety reminders

- › Muscles marked * are anatomical reference only: they are **not approved** for onabotA.
- › Adult ceiling of **400 U every 3 months**, ≤ 50 U per site and $\text{SCM} \leq 100$ U, even if the per-muscle ranges add up to more.
- › Scalenes, longissimus and the deep flexors benefit from EMG or ultrasound guidance.

Self-assessment questions

Question	Answer
A patient turns the chin to the left. Which SCM and which splenius are primary?	Right SCM (contralateral) and left splenius (ipsilateral).
Why is the SCM contralateral in torticollis but ipsilateral in laterocollis?	Because it rotates the head to the opposite side but tilts it toward its own side.

Question	Answer
CD-Track measures yaw 25° and roll 20°. What pattern does it report, and why?	Mixed: $25 / 20 = 1.25$, less than 1.5, so no component dominates.
Why does the app not suggest doses for anterocollis?	Greene 1990 has no template and Charles 2012 excluded isolated anterocollis; the deep flexors are not approved and the dysphagia risk is higher.
Which TWSTRS item makes you consider the levator scapulae?	D , shoulder elevation or anterior displacement.
Which muscle is primary only in retrocollis?	Semispinalis capitis.
What limitation does CD-Track have relative to the Col-Cap concept?	It measures total head orientation and does not separate head-on-neck from neck-on-trunk movement.

Final tips for Part 2

- › Four questions in order: **which plane?** → **toward which side of the patient?** → **which muscles on each side?** → **primaries only, or secondaries too?**
- › If you memorize one pair: **contralateral SCM + ipsilateral splenius** for rotation.
- › In laterocollis think "all on the same side"; in antero/retrocollis think "both sides".
- › Use the app's pattern as the starting point for the exam, not as the conclusion.